

L9:

Convex Optimization basics

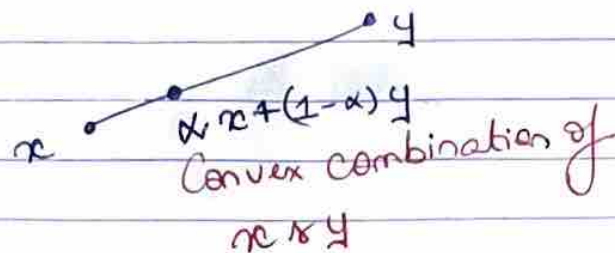
Today's agenda:

- Convex sets
- Convex functions
- Constrained convex optimization
- Lagrangian dual & KKT conditions

We will use this machinery next to tackle classification problems in supervised learning.

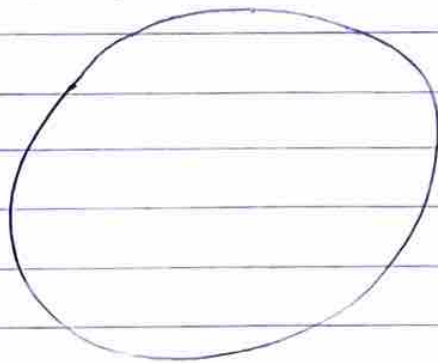
Convex sets

Say $x, y \in \mathbb{R}^n$, $\alpha \in [0, 1]$

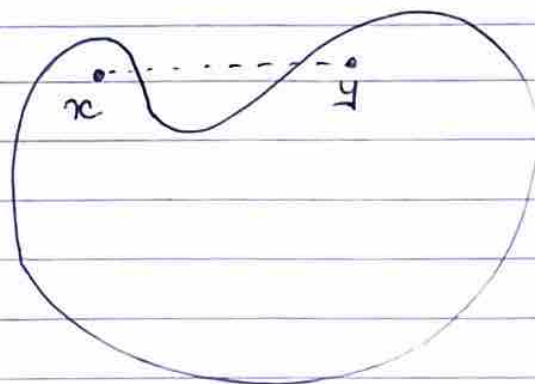


Def: A set $X \subseteq \mathbb{R}^n$ is convex if, for any $x, y \in X$, $\alpha \in [0, 1]$, $\alpha x + (1-\alpha)y \in X$.

{ In other words, the line segment joining any 2 points in X lies within X . }



CONVEX
SET



NOT A CONVEX SET

Eg Let A be $m \times n$ matrix, $b \in \mathbb{R}^m$ is a (column) vector.

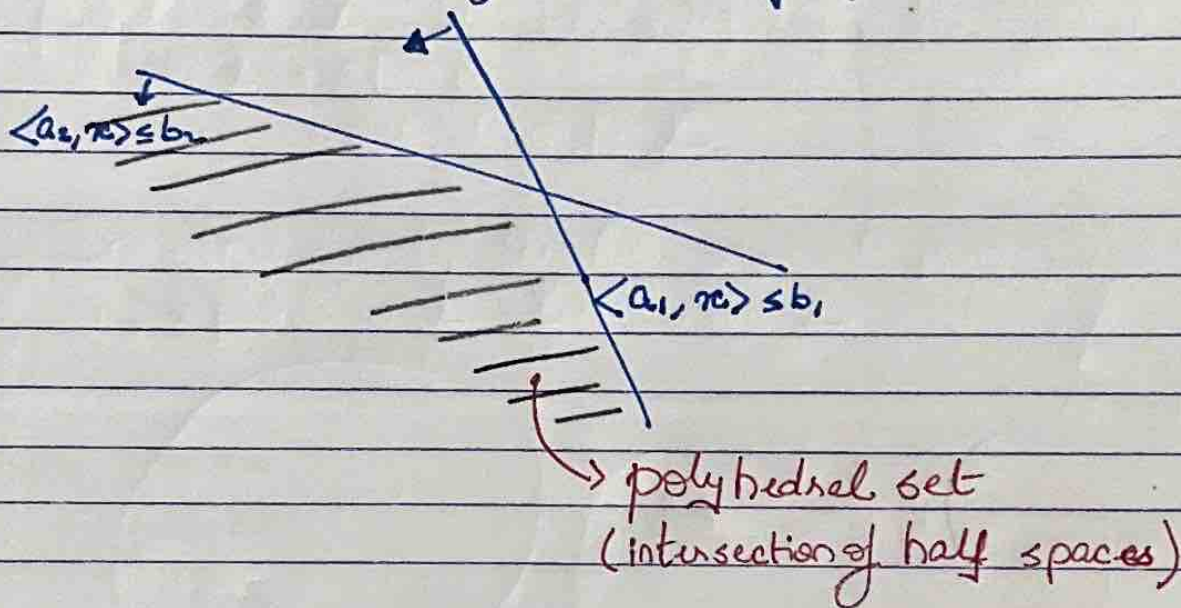
$\{x \in \mathbb{R}^n : Ax \leq b\}$ is convex.
pointwise

POLYHEDRAL SET

Let $A = \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix} \rightarrow \text{rows of } A$

Then $Ax \leq b \Leftrightarrow \begin{aligned} &\langle a_1, x \rangle \leq b_1 \\ &\langle a_2, x \rangle \leq b_2 \\ &\vdots \\ &\langle a_m, x \rangle \leq b_m \end{aligned}$

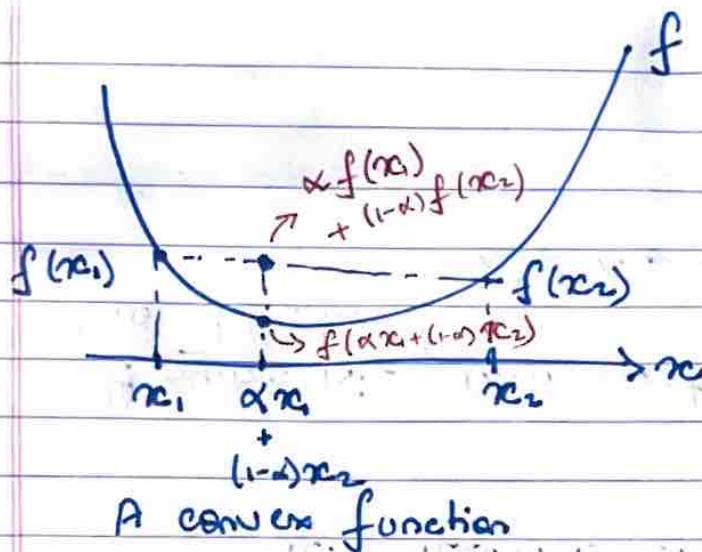
While $\langle a_i, x \rangle = b_i$ defines a hyperplane,
 $\langle a_i, x \rangle \leq b_i$ defines a half space



Convex functions

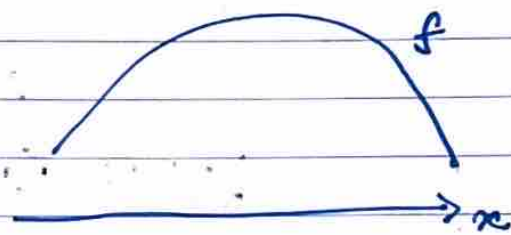
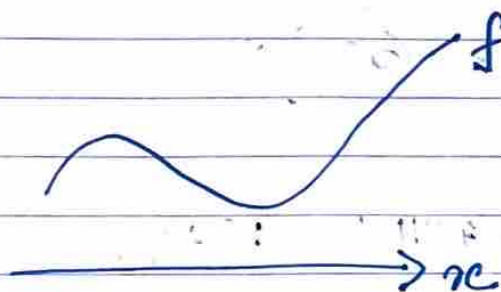
Def Say $X \subseteq \mathbb{R}^n$ is convex. A function $f: X \rightarrow \mathbb{R}$ is convex if for all $x_1, x_2 \in X$, $\alpha \in [0, 1]$,

$$f(\alpha x_1 + (1-\alpha)x_2) \leq \alpha f(x_1) + (1-\alpha)f(x_2)$$



★ It is "easy" to minimize convex functions!

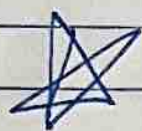
{ Graph of the function lies below the chord joining any 2 points on the graph }



★ It is easy to maximize concave functions

Def f is concave $\Leftrightarrow -f$ is convex.
 { Graph of function lies above the chord joining any 2 points on the graph. }

Recall



Say $f: \mathcal{X} \rightarrow \mathbb{R}$, where $\mathcal{X} \subseteq \mathbb{R}^n$.

Then at $x \in \mathcal{X}$,

$$\underbrace{\nabla f(x)}_{\text{Gradient of } f \text{ at } x} = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

Note: gradient is a vector in $\mathbb{R}^n$.

$$\underbrace{\nabla^2 f(x)}_{\text{Hessian of } f \text{ at } x} = \begin{bmatrix} \ddots & & \\ & \frac{\partial^2 f}{\partial x_i \partial x_j} & \\ & & \ddots \end{bmatrix}$$

Note: Hessian is an $n \times n$ matrix, it is symmetric.

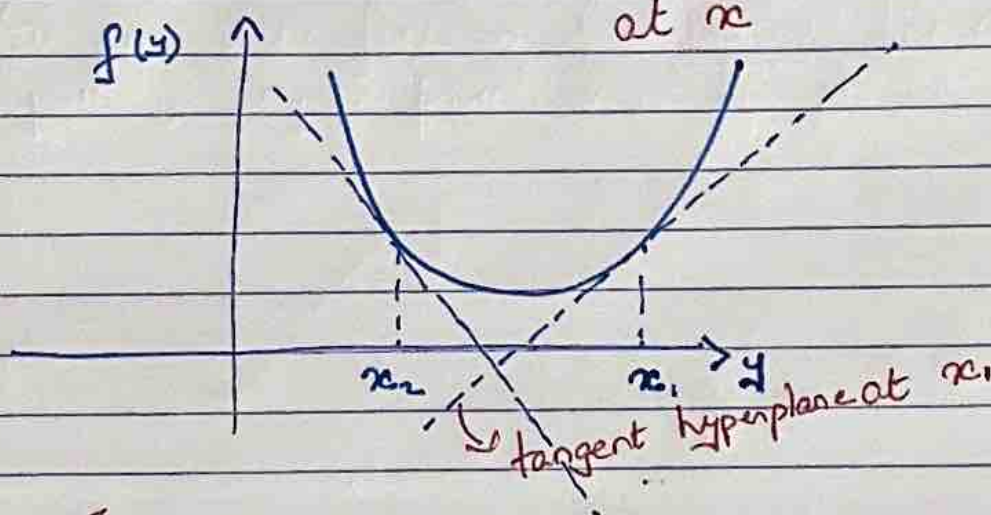


A symmetric $n \times n$ matrix J is positive semi-definite if $x^T J x \geq 0 \quad \forall x \in \mathbb{R}^n$.

Convexity^{also} has clear characterizations if f is differentiable.

Fact: Say $f: X \rightarrow \mathbb{R}$ is differentiable & X is convex. Then f is convex if and only if

$$f(y) \geq f(x) + \underbrace{\langle \nabla f(x), y - x \rangle}_{\text{tangent hyperplane at } x} \quad \forall x, y \in X$$



{ Graph of a convex function sits above any tangent hyperplane. }

Fact: Say $f: X \rightarrow \mathbb{R}$ is twice differentiable, and X is convex. Then f is convex iff $\underbrace{\nabla^2 f(x)}_{\text{Hessian}}$ is positive semi-definite $\forall x \in X$.

Eg: For a +ve semi definite matrix Σ ,

$x^T \Sigma x$ is convex.

Eg: $f(x) = \|x\|$ is a convex functions

Indeed

$$f(\alpha x_1 + (1-\alpha)x_2) = \|\alpha x_1 + (1-\alpha)x_2\|$$

$$\stackrel{\text{(Triangle inequality)}}{\leq} \|\alpha x_1\| + \|(1-\alpha)x_2\|$$

$$= \alpha \|x_1\| + (1-\alpha) \|x_2\|$$

Constrained convex optimization

$$(P) \quad \min_{x \in X} \underbrace{f(x)}_{\text{objective}} \quad \text{c.t.} \quad \underbrace{Ax \leq b}_{\substack{\downarrow \text{ } m \times n \quad \downarrow \text{ } m \times 1}} \quad \left. \vphantom{\min_{x \in X} f(x)} \right\} \text{CONSTRAINTS}$$

Annotations:
- X is a convex subset of $\mathbb{R}^n$.
- $f(x)$ is convex.
- $Ax \leq b$ are convex constraints.

- This is often referred to as a primal problem (P)
- Involves minimizing a convex objective f over a convex set $X \cap \{x: Ax \leq b\}$.
- Let p^* denote the optimal value of (P).

→ primal

Turns out: Solving (P) is equivalent to solving a sister optimization (D).

↓
dual

We now show how the dual is formulated.

The Lagrangian L is defined as follows:

$$\begin{aligned} L(x, \alpha) &= f(x) + \sum_i^m \alpha_i (a_i x - b_i) \\ &= f(x) + \alpha^T (Ax - b) \end{aligned}$$

Recall:
 $A = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix}$

Note: Constraints have been "moved" into the objective as "violation" penalties

Here,

- $\alpha_i \geq 0$ are Lagrange multipliers
- There is one Lagrange multiplier for each constraint in (P).

Def: The (Lagrange) dual objective F is defined as:

$$F(\alpha) = \inf_{x \in X} L(x, \alpha)$$

Think of as
min

- Turns out F is a concave function of the Lagrange multipliers.

The Dual optimization corresponding to (P) is:

$$\max_{\alpha \in \mathbb{R}_+^m} F(\alpha) \quad (D)$$

- (D) maximizes a ~~convex~~ concave function over a convex constraint set
- The optimization variables of (D) are Lagrange multipliers corresponding to the constraints of (P).
- Let d^* denote the optimal value of (D).
Turns out: $p^* = d^*$
↳ Strong duality
- In practice, solving (D) is equivalent to solving (P).
But it is sometimes easier and/or more insightful than (P).

Theorem: x^* solves (P) if and only if

$$\exists \alpha^* \in \mathbb{R}_+^m \text{ s.t.}$$

KKT Conditions

$$\begin{cases} \nabla_x L(x^*, \alpha^*) = \nabla_x f(x^*) + \sum \alpha_i^* a_i = 0 \\ Ax^* \leq b \\ \alpha_i^* (a_i x^* - b_i) = 0 \quad \forall 1 \leq i \leq m \end{cases}$$

$\left\{ \begin{array}{l} x^* \text{ minimizes } L \text{ fixing } \alpha^* \\ \text{Primal feasibility} \end{array} \right\}$

$\hookrightarrow$ Complementary slackness

In this case, α^* solves (D).

Moral: $\text{Primal (P)} \iff \text{Dual (D)}$.

Solving either is equivalent to finding (x^*, α^*) satisfying the KKT conditions.

Eg: Consider the following constrained convex optimization.

$$\begin{aligned} \min. \\ (x_1, x_2) \in \mathbb{R}^2 \end{aligned}$$

$$x_1^2 + 4x_2^2$$

$$\text{s.t. } x_1 + x_2 \leq -1$$

Then

$$\underset{\text{Lagrangian}}{\mathcal{L}} = x_1^2 + 4x_2^2 + \underset{\text{Lagrange multiplier}}{\alpha}(x_1 + x_2 + 1)$$

Minimize Minimizing $\mathcal{L}$ w.r.t. x_1, x_2

$\Downarrow$

$$x_1^* = -\alpha/2, \quad x_2^* = -\alpha/8$$

$$\Rightarrow \underset{\text{Dual objective}}{F(\alpha)} = \left(-\frac{\alpha}{2}\right)^2 + 4\left(-\frac{\alpha}{8}\right)^2 + \alpha\left(-\frac{\alpha}{2} - \frac{\alpha}{8} + 1\right)$$

$$= \alpha\left(1 - \frac{5\alpha}{16}\right)$$

$$\Rightarrow \text{Dual is: } \max_{\alpha \geq 0} \alpha\left(1 - \frac{5\alpha}{16}\right)$$

$$\Rightarrow \alpha^* = 8/5$$

$\left\{ \text{Dual solution} \right\}$

$$\Rightarrow (x_1^*, x_2^*) = \left(-\frac{4}{5}, -\frac{1}{5}\right)$$

(by KKT)

$\left\{ \text{Primal sol} \right\}$